

KATHMANDU UNIVERSITY
Department of Computer Science and Engineering
Dhulikhel, Kavre



A Project Proposal
On
“KU InfoBot: A Rule-Based Chatbot for KU”

[Code No.: ENGG 102]

(For partial fulfillment of Year I / Semester II in Computer Engineering)

Submitted by:

Smriti Niroula (Roll No: 17)

Prayag Parajuli (Roll No: 30)

Aislinn Paudel (Roll No: 32)

Ajesh Pokhrel (Roll No: 40)

Prashis Prasai (Roll No: 45)

Submitted to:

Suman Shrestha

Department of Computer Science and Engineering

April 27, 2026

Abstract

Chatbot systems are increasingly used in educational institutions to provide quick and automated responses to user queries. However, accessing organized and accurate university information remains challenging for students due to scattered sources.

The main aim of this project is to develop a chatbot system that provides structured information about Kathmandu University. The system is designed to assist users in obtaining details related to courses, admission procedures, fee structures, and facilities.

The chatbot will be implemented using C and C++ programming language with a rule-based approach. It will utilize concept of Object Oriented Programming (OOP) and basic programming constructs such as conditional statements, loops, and string handling to process user inputs and generate appropriate responses.

The expected outcome of this project is a functional and user-friendly chatbot capable of responding to predefined queries efficiently. It will improve accessibility of information and reduce the time required for manual searching.

In conclusion, the project demonstrates the practical application of fundamental programming concepts in solving real-world problems. It is recommended that future work should incorporate artificial intelligence techniques to enhance the chatbot's capability in understanding complex user queries.

Keywords: *Chatbot, C, C++, University Information, Automation, InfoBot*

Contents

Abstract.....	i
List of Figures	iii
Abbreviations.....	iv
Chapter 1: Introduction	1
1.1 Background.....	1
1.2 Objectives	2
1.3 Motivation and Significance	2
1.4 Expected Outcomes.....	3
Chapter 2: Related Works	4
Chapter 3: Procedure and Methods	5
Chapter 4: System Requirement Specifications	7
4.1 Software Specifications.....	7
4.2 Hardware Specifications.....	8
Chapter 5: Project Planning and Scheduling	9
5.1 Tasks.....	10
References	11

List of Figures

Figure 3.1: Flowchart of InfoBot Architecture.....	6
Figure 4.1: Use-Case Diagram of KU InfoBot.....	7
Figure 5.1: Gantt Chart of Project Schedule.....	9

Abbreviations

AI: Artificial Intelligence

GCC: GNU Compiler Collection

IBM: International Business Machine

I/O: Input/Output

NLP: Natural Language Processing

OOP: Object Oriented Programming

Chapter 1: Introduction

Chatbots are computer-based systems that simulate human conversation and provide automatic responses to user queries. Their use has increased in fields like education, healthcare, and business. Many educational institutions use chatbots to provide quick information about courses, admissions, and schedules. Advanced chatbots use Artificial Intelligence (AI) and Natural Language Processing (NLP), but they require complex technology and are not suitable for beginner projects. Therefore, this project develops a simple rule-based chatbot using C and C++ to provide Kathmandu University information efficiently.

1.1 Background

Chatbots are computer-based systems designed to simulate human conversation and provide automated responses to user queries. In recent years, there has been significant growth in the use of chatbots across various sectors such as education, healthcare, and business. Educational institutions, in particular, are increasingly adopting chatbot systems to provide instant information to students regarding courses, admissions, schedules, and other academic services.

Recent developments in chatbot technology involve the integration of AI and NLP, which allow systems to understand and respond to complex human language. Many universities have implemented web-based or mobile chatbot systems that can handle a wide range of queries. These systems improve accessibility, reduce administrative workload, and enhance user experience by providing 24/7 support.

However, such advanced systems require sophisticated technologies, large datasets, and complex algorithms, which are not suitable for beginner-level projects. Additionally, existing simple systems often lack structure, flexibility, and clarity in handling user inputs. Therefore, there is a need for a simplified, structured chatbot

system that can efficiently provide university-related information using basic programming techniques. This project addresses that need by developing a rule-based chatbot using C and C++.

1.2 Objectives

The main objectives of this project are:

- To design and develop a chatbot using C and C++ programming language
- To provide structured information about Kathmandu University
- To implement a user-friendly console-based interaction system
- To demonstrate the practical application of basic programming concepts

1.3 Motivation and Significance

The motivation behind this project arises from the difficulty students often face in accessing accurate and organized university information. Students frequently need information about courses, admissions, fees, and facilities, but searching through multiple sources can be time-consuming and inefficient.

This project aims to address these issues by providing a centralized system that delivers quick and reliable responses. Unlike advanced AI-based chatbots, this system uses a rule-based approach, making it simpler and more suitable for first-year students. It eliminates the complexity of NLP while still maintaining effective interaction.

The significance of this project lies in its ability to bridge theoretical knowledge and practical implementation. It enhances programming skills, logical thinking, and problem-solving abilities.

1.4 Expected Outcomes

A functional rule-based chatbot system implemented using C and C++ is developed to achieve the objective of providing structured information through an interactive interface. The system is capable of accepting user queries, processing input using preprocessing techniques such as typo correction and synonym mapping, and performing basic intent recognition through keyword-based classification. It generates appropriate responses based on predefined rules, ensuring accurate and consistent information delivery. The project fulfills its objectives by demonstrating the application of fundamental programming concepts, including control structures, string handling, and modular design. Additionally, it achieves the goal of developing a simple, efficient, and user-friendly system for information retrieval. Overall, the outcome reflects a successful implementation of a structured chatbot system that meets the defined objectives while maintaining reliability and ease of use.

Chapter 2: Related Works

Chatbots have been widely adopted in educational institutions to automate information delivery and improve accessibility. International Business Machine (IBM) defines chatbots as systems that simulate human conversation to provide automated assistance, while GeeksforGeeks distinguishes between rule-based and AI-driven approaches. Rule-based systems are predictable and simple, whereas AI-based systems offer flexibility but require complex algorithms and large datasets.

In Nepal, practical implementations of educational chatbots have already been demonstrated. For example, Sigdel, Shrestha, and Neupane (2024) developed a **College Chatbot Using RASA** at Kathmandu Engineering College, which utilized NLP and machine learning to answer student queries with high accuracy. Similarly, Nepal's **Conversational AI Research Agenda (2025)** emphasizes the importance of chatbots in education, highlighting challenges in low-resource languages like Nepali and proposing frameworks such as NepBERTa to improve performance.

Compared to these systems, the proposed project uses a rule-based chatbot developed in C and C++, focusing on simplicity and ease of implementation. The system includes typo correction, synonym handling, and basic intent recognition to improve input processing. It provides structured and accurate information about Kathmandu University. Although limited in handling complex queries, it effectively delivers reliable and user-friendly information retrieval.

Chapter 3: Procedure and Methods

Initially, data collection will be performed to gather relevant information about Kathmandu University. This includes details such as courses offered, admission procedures, fee structures, available facilities etc. The collected information will be obtained from reliable sources such as the official university website and related documents. After collection, the data will undergo a digitization process, where the information will be organized and converted into a structured format suitable for implementation within the chatbot system.

Following this, requirement analysis will be conducted to determine the specific functionalities of the chatbot based on the collected data. Based on these requirements, the system design will be developed, including the interaction flow between the user and the chatbot.

The implementation phase will involve coding the chatbot using C and C++ programming language. Basic programming constructs such as loops, conditional statements, and string handling will be used to process user input and generate appropriate responses. A rule-based approach will be implemented where specific keywords are matched with predefined outputs.

Finally, testing and debugging will be carried out to ensure that the chatbot responds accurately to various user inputs. The system will then be refined and properly documented for final submission.

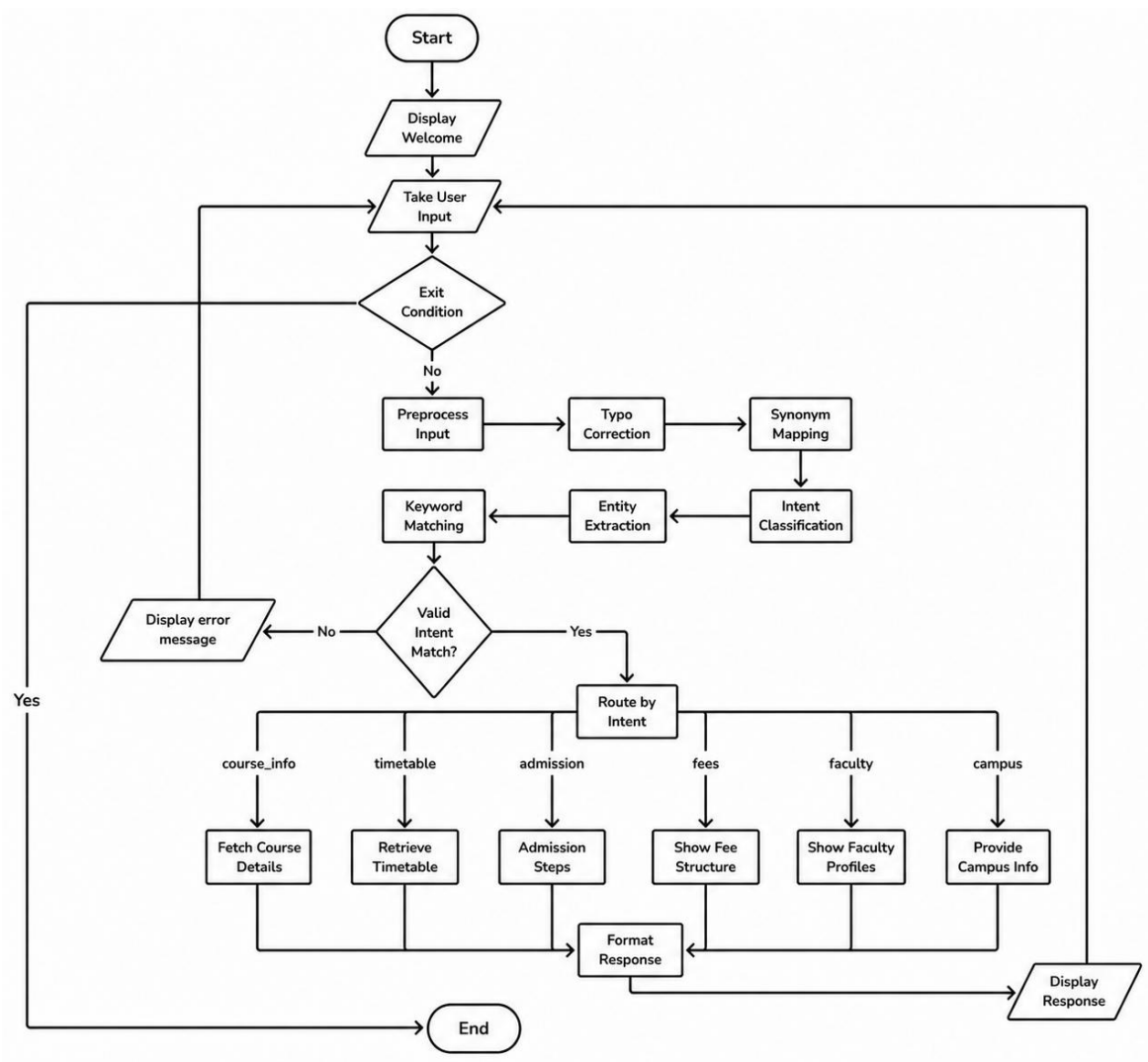


Figure 3.1: Flowchart of InfoBot Architecture

Chapter 4: System Requirement Specifications

In order to build and operate the KU InfoBot, the project requires a set of defined hardware and software specifications. These specifications serve as the foundation for ensuring smooth functionality, reliable performance, and effective interaction between the user and the system.

4.1 Software Specifications

- **Programming Language:** C/C++
- **Compiler:** GCC (GNU Compiler Collection)
- **Framework and Libraries:** Qt, Standard C/C++ libraries
- **Development Tools:** Visual Studio Code, Git, Github
- **Operating System:** Windows / Linux / MacOS
- **Build System:** CMake

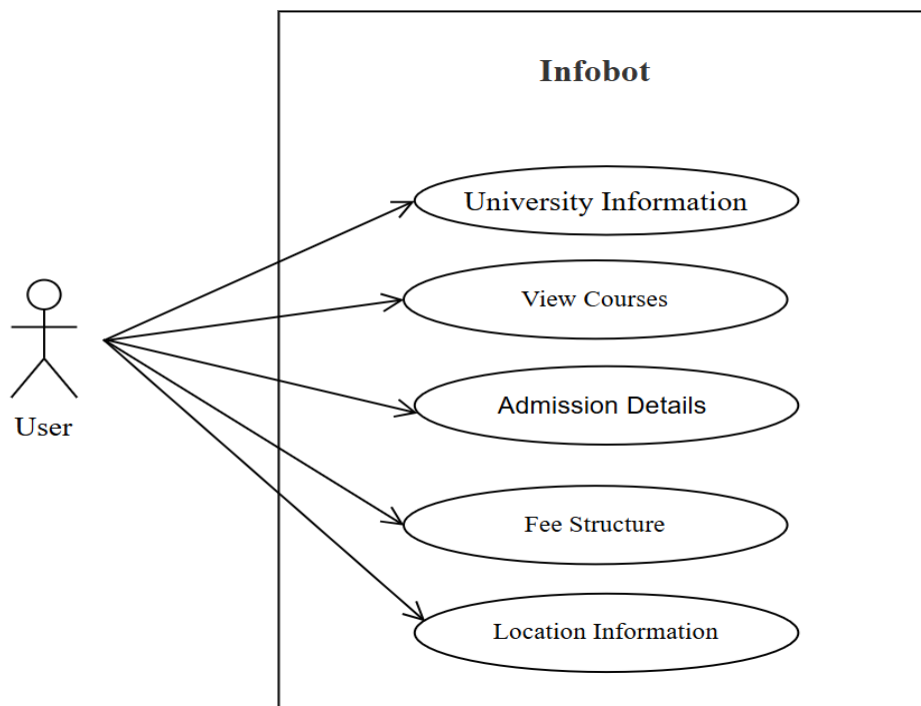


Figure 4.1: Use-Case Diagram of KU

4.2 Hardware Specifications

The hardware requirements for the system are minimal. The chatbot can run on any basic computer system with the following configuration:

- Processor: Intel i3 or above
- RAM: 2 GB or more
- Stable internet connectivity
- I/O Devices: Keyboard and monitor

Chapter 5: Project Planning and Scheduling

To carry out the KU InfoBot project in an organized and timely manner, a clear plan and schedule are essential. Proper structuring of tasks, allocation of time, and systematic tracking of progress ensure that each stage of development is completed efficiently. By outlining the sequence of activities from problem identification and requirement gathering to final submission, the project establishes a roadmap that guides the team toward successful completion.



Figure 5.1: Gantt Chart of Project Schedule

The project will begin with problem identification and requirement gathering, followed by system design and data collection along with digitization. The coding and implementation phase will take the longest duration, followed by testing, documentation, review, and final submission.

5.1 Tasks

The major tasks involved in the project are:

- Problem Identification and Requirement Analysis
- Data Collection and Digitization
- System Design
- GUI Design using Qt Framework
- Coding and Implementation using C/C++
- Integration, Testing, and Debugging
- Documentation, Final Review and Submission

References

- GeeksforGeeks. (n.d.). *Chatbot Theory Explained!* Retrieved from GeeksforGeeks: <https://www.geeksforgeeks.org/blogs/chatbot-theory-explained/>
- IBM. (n.d.). *Chatbots*. Retrieved from IBM: <https://www.ibm.com/think/topics/chatbots>
- Sigdel, Y., Shrestha, S., & Neupane, N. (2024). College chatbot using RASA. *KEC Journal of Science and Engineering*, 8(1), 38-43.
doi:<https://doi.org/10.3126/kjse.v8i1.69263>